

One-Year AI / ML / Data Science / Generative AI / Agentic AI Roadmap

Goal: Build strong foundations in Data Science, Machine Learning, Deep Learning, Generative AI, and Agentic AI within one year while maintaining DSA practice and project work.

Month 1: NumPy

- Arrays, indexing, slicing, reshaping
- Mathematical operations and statistics functions
- Weekend project: Student Marks Analyzer

Month 2: Pandas

- Series, DataFrames, CSV files
- Filtering, sorting, missing values, groupby
- Projects: IPL / Netflix Data Analysis

Month 3: Visualization + Statistics

- Matplotlib and Seaborn
- Mean, median, mode, variance, probability
- Project: Sales Dashboard

Month 4: SQL

- SELECT, WHERE, GROUP BY, HAVING
- JOINS and subqueries
- Project: Student Database Analysis

Month 5: Math for ML

- Linear Algebra
- Probability
- Basic Calculus for ML

Month 6: Machine Learning Foundations

- Linear Regression
- Logistic Regression
- KNN, Decision Trees, Random Forest

Month 7: Advanced Machine Learning

- Feature Engineering
- Hyperparameter Tuning
- PCA and Ensemble Methods

Month 8: Deep Learning

- Neural Networks
- Backpropagation
- PyTorch Basics

Month 9: Computer Vision and NLP

- CNNs
- Text Processing
- Sentiment Analysis Project

Month 10: Generative AI

- Transformers
- Embeddings
- Prompt Engineering

Month 11: RAG and LLM Applications

- Vector Databases
- RAG Systems
- PDF Chatbot Project

Month 12: Agentic AI + Deployment

- LangGraph
- CrewAI
- FastAPI, Docker, Deployment

Weekly Study Structure

Monday–Friday

- 45 minutes DSA
- 60–90 minutes AI/ML learning
- 15–30 minutes revision and notes

Saturday

- 2–3 hours project building

Sunday

- Finish project
- Push work to GitHub
- Weekly revision

Target Portfolio Projects

- Student Marks Analyzer
- IPL Data Analysis
- Sales Dashboard
- House Price Predictor
- Spam Classifier
- Image Classifier
- Resume Analyzer
- PDF Chatbot
- AI Research Agent